Part 1: Design Proposal for a Wearable Electromyography Device

ELEN4006

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Abstract – This paper presents the high-level design of an electromyography (EMG) device, for capturing muscle activity data to support gesture identification for a bionic hand used by amputees. The design follows guidelines from the Surface Electromyography for the Non-Invasive Assessment of Muscles (SENIAM) project and established biomedical standards for EMG instrumentation. The design includes electrodes to measure electric muscle potential, a differential amplifier to eliminate common noise from the electrodes, and a band pass filter to remove interference from distant muscles. The processed data is then analysed and presented for analysis by a medical professional. A trusted database of raw EMG data from amputees is used to visualise expected signal patterns and validate the design proposal. The results indicate that the design is a viable solution, providing a foundation for further development and implementation.

Keywords – Electromyography (EMG), SENIAM

1. Introduction

Limb amputation impairs physical functionality and mobility, with the leading causes being diabetes, vascular disease and trauma [1]. In 2017 about 57.7 million people lived with limb amputations and McDonald *et al* [1] estimated that 75,850 prosthetics were needed for all trauma related amputations. Prosthetic advancements have improved dexterity and functionality by using multiple degrees of freedom, enabled by feature classification of electric potential signals from muscle contractions[2], [3], [4]. However current systems rely on stationary EMG data collection limiting real world use[4], [5]. This paper proposes a design for a wearable EMG data acquisition system to enable prosthetic control. Section 2 covers the measurement application, section 3 analyses the measured variable and section 4 highlights the design specifications. Section 5 and 6 review existing solutions and present the high-level design respectively. Section 7 discusses sensor selection and section 8 provides the conclusion.

2. Description of measurement application

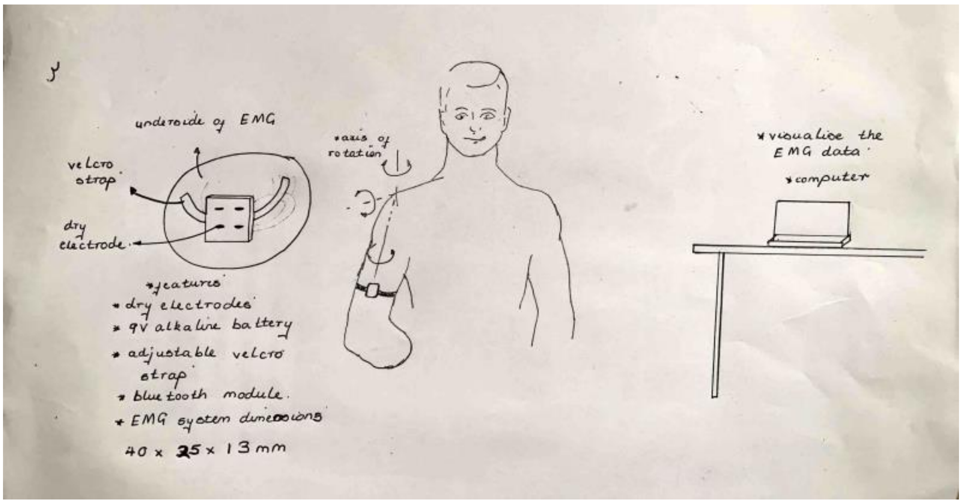

Figure 1: Visual representation of the measuring device.

The proposed EMG device will feature an adjustable Velcro strap for secure forearm placement. As shown by figure 1, electrode placement will follow SENIAM standards for high quality signal acquisition [6]. A compact housing case will contain the electrodes, differential amplifier, and bandpass filter to minimise noise and movement artefacts[7], a consequence of the dry electrodes high impedance [8]. A Bluetooth module will wirelessly transmit captured data to a computer for visualisation for a medical professional. The device will function in a dynamic environment, that includes walking and light physical activity, minor skin preparation using alcohol swabs is necessary for optimal signal acquisition. The wireless data transmission should remain stable during movement, it is paramount that the device minimize noise caused by movement and the battery may require periodic placements.

3. Analysis of measuring variable

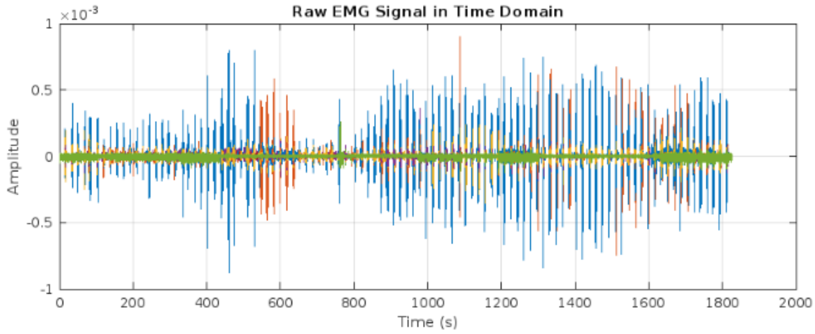

Figure 2: Time domain analysis of raw EMG data for an amputee [9].

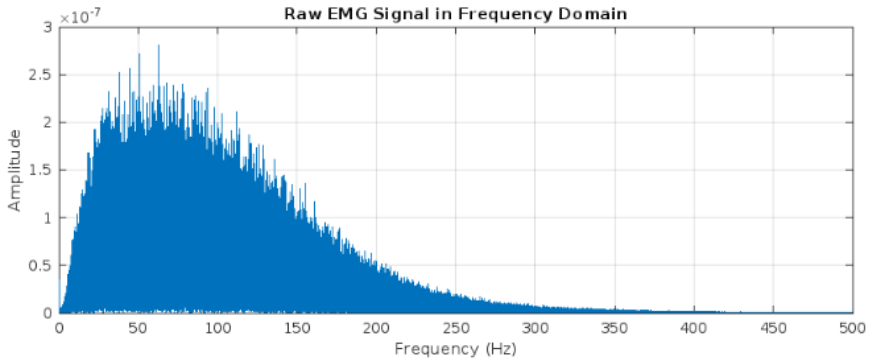

Figure 3 : Frequency analysis of raw EMG data from Ninapro database [9].

Figure 2 shows the digitised raw EMG data from the Ninapro database [9], with a maximum amplitude of 0.89 mV, within the expected range of 10 mV peak amplitude as per literature [3]. The red line represents baseline noise from power line interference at 50 Hz and motion artefacts [10]. Figure 3 presents the frequency analysis of the raw EMG data using a Fast Fourier Transform (FFT) algorithm at a sampling rate of 1000Hz in MATLAB. Muscle activity primarily occurs between 50 Hz and 150 Hz [11], shown by the high amplitudes within this region. As frequency increases there is a gradual loss of amplitude approaching zero at 500 Hz due to an increase in distant muscle contributions [10]. This sets the EMG bandwidth range as 0 – 500 Hz, setting a sample rate of 1000Hz for the FFT based on Nyquist principles[7].

4. Design specification

The journal papers [7], [10], [12], [13] provide a framework for the standards that allow a EMG to provide the most accurate results. This framework allows for the definition of the following design specifications provided in table 1.

Table 1: Design specifications for the proposed EMG device.

Specifications	Range	Justification
Input Values	25μV – 10 mV	The amplitude of the EMG signal varies from a range of μV to mV, with literature supporting values from 25 μV to 10 mV [3].
Signal to Noise Ratio	20 – 40 dB	Based on the recommended range that allows for reliable EMG data [14].
Input Impedance	1 – 10 MΩ	This value is the suggested range that encompasses dry and wet electrodes [15].
Bandwidth	10Hz –700Hz	The EMG signal bandwidth is set based on existing literature[7] ,[10]. Most muscle activity signals are concentrated between 10 Hz and 500 Hz [7] , but the upper limit ensures finer signal detail while balancing noise rejection.
Power Supply	± 7V	The Arduino that will be used for signal processing has a voltage requirement of 7 – 20 V for optimum performance [16].
Common Mode Rejection Ratio	100 –120 dB	This is a measure of how well the differential amplifier can suppress interference and noise that is common to the two-channel electrodes. SENIAM recommend a value greater than 95dB as acceptable with 120 dB the highest achievable value [17].

5. Review of existing solutions

Table 2: Review of existing solutions.

Criteria	Wearable EMG shirt [18].	Wearable EMG-based for Human Computer Interface[19].	Portable EMG using electrode arrays [20].
Description	Shirt fitted with textile sensors on the inside that capture EMG data and send data to computer for display and analysis.	Wearable EMG sleeve for human computer interface for electric powered wheel chairs.	EMG data acquisition using electrode arrays for high density surface EMG.
Electrode used	Medtex P-130 conductive fabric for dry textile electrodes.	13E125 active surface electrode.	Kapton grid of 32 Ag electrodes.
Data Transmission	HC-05 Bluetooth module connected to a personal computer.	Bluetooth module.	Wi-Fi connectivity.
Critical Analysis	A good possible solution.	This is limited to only wheelchair patients with spinal injuries and above elbow amputation. It is also stramineous weighing in 140g.	Applicable for this paper as a viable solution. The only limitation is the router signal availability.

6. Proposed high-level design.

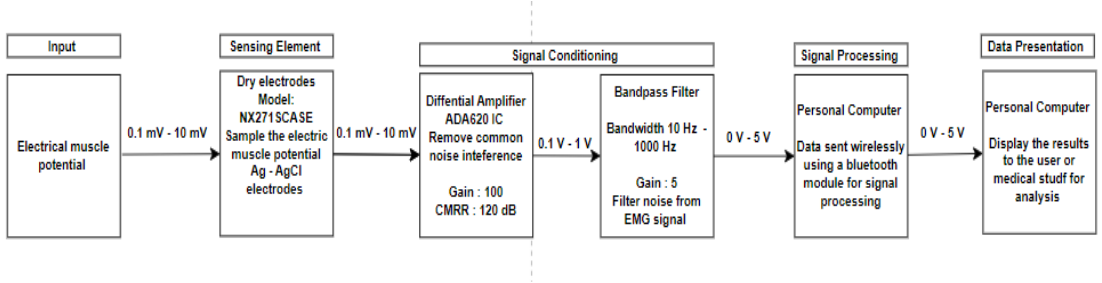

Figure 4: High level design of proposed EMG measuring device.

7. Sensor selection

The use of electrodes allows for the measuring device to show a faithful representation of the muscle fibre potential. Below is a table for the comparison of traditional EMG electrodes.

Table 3: Electrode comparison for application to high level design.

Electrode Type	Pros	Cons
Needle Model: EL450	Direct recording from muscle fibres producing a very high signal quality. Low noise and high spatial resolution. Less affected by motion artefacts.	Invasive and can be uncomfortable for the user and induce pain. Risk of infection or tissue damage. Not suitable for long term use or wearable
Dry Electrode Model: NX271SCASE	No need for skin preparation or conductive gel. More convenient for long term and wearable use. Easier to apply and remove.	High skin electrode impedance may deteriorate signal to noise ratio. More susceptible to motion artefacts. Lower signal quality compared to wet electrodes.
Gel Electrode Model: 76642R	The use of Ag-AgCl introduces less electrical noise compared to metallic material allowing for higher quality EMG signal quality[7]. Made to be very lightweight , less than 1gram [7] and have a reusable version.	Reusable gel electrodes result in signal quality degradation over multiple uses[7]. Gel can dry out over time, reducing signal quality. Potential for skin irritation from the gel.

The NX271SCASE dry electrode, is the best choice for this prosthetic application due to their suitability for daily use and mobility. The drawbacks, such as higher impedance [8], can be mitigated by using a preamplifier with a input impedance exceeding the skin electrode range.

Conclusion

The proposed design works as it abides by SENIAM standards and a implementation will be pursued.

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